Sindh Science Learning Portal

Class 5 · Science

Chapter 9

Soils



Learn what soil is made of, its properties and types, and which living things live in it.

Following the Sindh Textbook Board curriculum

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CHAPTER 9

Soils

What we will learn Learn what soil is made of, its properties and types, and which living things live in it.

1 What is soil made of?

The soft top layer on the surface of the Earth is called **soil**. Plants grow in soil.

Soil is made from **rocks**. Heat, cold, rain and wind break rocks into tiny pieces over thousands of years.

Soil contains:

- **Mineral particles**: grains of sand, silt and clay.
- **Humus**: rotted leaves, plants and dead animals. Humus makes soil rich.
- **Air** and **water**, and many **living things**.



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2 Properties of soil

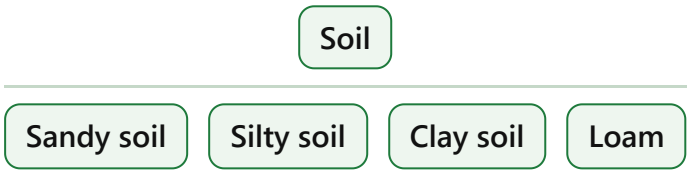
- **Colour**: soil can be red, brown, black or grey. Black and dark brown soils usually contain a lot of humus.
- **Texture**: rub some wet soil between your fingers. Sand feels **gritty**, silt feels **smooth and silky**, and clay feels **sticky**.
- **Compaction**: in tightly packed, hard soil, little air and water can get in and roots grow with difficulty.
- **Holding water and drainage**: some soils let water run through quickly, while others hold it for a long time. Good soil holds enough water and lets the extra water drain away.

3 Types of soil

- **Sandy soil**: large, gritty grains. Water drains through it very fast, so it holds little water and few nutrients. It is found in deserts like the Thar.
- **Silty soil**: fine, smooth grains. It holds water well and is fertile. It is found along the banks of the River Indus.
- **Clay soil**: very tiny grains. It is sticky when wet and hard when dry. It holds a lot of water, so water stands on it.
- **Loam**: a mixture of sand, silt, clay and humus. It holds the right amount of air and water, so it is the **best soil for plants**.

Grain size: sand is the largest, then silt, and clay grains are the smallest.

Diagram: the parts and what they do



Soil (مٹي · مٹی)
The soft layer made from broken rock, with mineral grains, humus, air, water and living things.

Sandy soil (واريءَ واري مٹي · ریتلی مٹی)
Large gritty grains. Water drains away fast. Found in deserts.

Silty soil (گاد واري مٹي · گادی مٹی)
Smooth silky grains. Holds water well and is fertile. Found along rivers.

Clay soil (چيڪي مٹي · چکنی مٹی)
The tiniest grains. Sticky when wet, hard when dry. Holds a lot of water.

Loam (لوم · میرا مٹی)
A mix of sand, silt, clay and humus. The best soil for plants.

4 Life in the soil and soil erosion

Soil is home to many living things. A single spoonful of soil can hold millions of microbes.

- **Bacteria and fungi** rot dead plants and animals and return **nutrients** to the soil. They are called **decomposers**.
- **Earthworms** dig tunnels that let air and water into the soil. They mix the soil and make it rich.
- Ants, insects and moles also live in the soil.

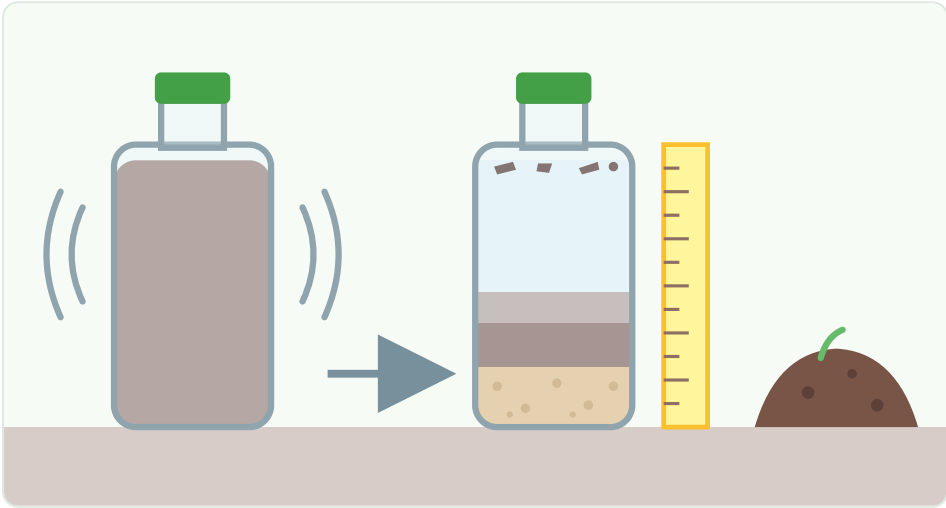
Running water and strong wind can carry away the rich top layer of soil. This is called **soil erosion**. Planting trees and grass helps, because roots hold the soil in place.

Key words

English	سنڌي	اردو
soil	مٹي	مٹی

English	سنڌي	اردو
sand	ريتي	ریت
silt	گاد	گاد
clay	چيڪي مٽي	چکنی مٹی
loam	لوم	میرا مٹی
humus	هيومس	ہیومس
minerals	معدنيات	معدنیات
decomposer	سڙائيندڙ	تحلیل کار
earthworm	ڪنڊيو	کیچوا
erosion	مٽيءَ جو کاتو	مٹی کا کٹاؤ

Experiment: What is in soil?



You need

- A clear plastic bottle or jar with a lid
- Two handfuls of garden soil
- Water
- A ruler

Steps

1. Fill one third of the bottle with soil.
2. Add water until the bottle is almost full, and close the lid tightly.
3. Shake the bottle well for one minute.
4. Leave the bottle still for one day.
5. Look at the layers and measure each layer with the ruler.

Safety: Use a plastic bottle instead of glass. Do not put soil in your mouth, and wash your hands with soap afterwards.

What did we learn? The soil settled in layers: the heaviest sand at the bottom, silt above it and fine clay on top. Bits of rotted leaves (humus) floated on the water. This shows that soil is a mixture of many things.

Practice questions

Circle the correct answer

1. **What is soil made from?**

   (a) Broken-down rocks **(b)** Plastic
   (c) Only water **(d)** Clouds

2. **What is the part of soil made from rotted leaves and dead animals called?**

   (a) Sand **(b)** Humus
   (c) Clay **(d)** Rock

3. **How does wet clay feel when you rub it between your fingers?**

   (a) Gritty **(b)** Sticky
   (c) Hot **(d)** Dry

4. **What does dark black soil usually have a lot of?**

   (a) Humus **(b)** Plastic
   (c) Salt **(d)** Glass

5. **Which soil is the best for plants?**

   (a) Sandy soil **(b)** Clay soil
   (c) Loam **(d)** Rock

6. **Which soil lets water drain away fastest?**

   (a) Clay soil **(b)** Sandy soil
   (c) Loam **(d)** Silty soil

7. **Which soil has the smallest grains?**

   (a) Sand **(b)** Silt
   (c) Clay **(d)** All are equal

8. **How do earthworms help the soil?**

   (a) They dig tunnels that let air and water in **(b)** They make the soil hard
   (c) They eat all the plants **(d)** They dry up the water

9. **What is it called when running water and wind carry away topsoil?**

   (a) Germination **(b)** Pollution
   (c) Soil erosion **(d)** Recycling

Write true or false

10. **Soil also contains air and water.** _____

11. **Roots grow easily in hard, tightly packed soil.** _____

12. **Clay soil becomes hard when it dries.** _____

13. **No living things live in soil.** _____

14. **Planting trees reduces soil erosion.** _____

Match the columns

15. **Match each soil with what it is like.**

1. Sandy soil		(a) Water stands on it
2. Clay soil		(b) Best for plants
3. Loam		(c) Drains fast

Match the columns

16. **Match each living thing with its job in the soil.**

1. Earthworm		(a) Rot dead things
2. Bacteria		(b) Digs tunnels

Worksheet: Soils

Name: _____ Date: _____

A. Fill in the blanks

Word box: Earthworms · Loam · erosion · humus · rocks

1. Soil is made from broken-down _____.
2. The rich part of soil made from rotted leaves is called _____.
3. _____ is the best soil for plants.
4. _____ dig tunnels that let air into the soil.
5. Soil being carried away by water and wind is called _____.

B. Put ✓ for true and ✗ for false

1. ☐ Sand grains are bigger than clay grains.
2. ☐ Sandy soil holds a lot of water.
3. ☐ Decomposers break down dead things into the soil.
4. ☐ Cutting down trees reduces soil erosion.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Sand	( ) Sticky
2. Silt	( ) Rotted leaves
3. Clay	( ) Gritty
4. Humus	( ) Smooth and silky

D. Answer in short

1. Name four things found in soil.

2. Why is loam the best soil for plants?

3. How can we stop soil erosion?

E. Draw and label

Draw the soil-in-a-bottle test and label the layers of sand, silt, clay and humus.

FOR TEACHERS

Answer key

Chapter 9: Soils

Practice questions

1. (a) Broken-down rocks
2. (b) Humus
3. (b) Sticky
4. (a) Humus
5. (c) Loam
6. (b) Sandy soil
7. (c) Clay
8. (a) They dig tunnels that let air and water in
9. (c) Soil erosion
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. rocks · 2. humus · 3. Loam · 4. Earthworms · 5. erosion

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Sticky · (4) Rotted leaves · (1) Gritty · (2) Smooth and silky

D. Answer in short:

1. Mineral grains, humus, air and water (and living things).
2. It has sand, silt, clay and humus, and holds the right amount of air and water.
3. By planting trees and grass, because roots hold the soil in place.

Picture credits

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